

Haedo Cho

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EDUCATION

Harvard University

Ph.D., in Electrical Engineering

Thesis Advisor: Professor Patrick Slade

Sep. 2020 - Aug. 2026 (Anticipated)

Korea Advanced Institute of Science and Technology

M.S., in Mechanical Engineering

Aug. 2016

Inha University, Incheon, Korea

B.S with Honors, *Summa Cum Laude*, Mechanical Engineering

Aug. 2014

RESEARCH EXPERIENCE

Harvard University, [Slade Lab](#)

Graduate Researcher with Prof. Patrick Slade

Jan. 2023 – Present

- **Smartphone-based real-world energy expenditure estimation**

- Developed a biomechanically inspired, data-driven model to estimate energy expenditure from smartphone data.
- Built a scalable AWS pipeline for storing and processing smartphone sensor data.
- Designed and conducted Harvard IRB-approved human-subject studies in real-world settings.

Harvard University, [Biodesign Lab](#)

Graduate Researcher with Prof. Conor Walsh

Sep. 2020 – Jun. 2022

- **IMU-based tracking system for strength training**

- Developed a portable embedded Linux system with real-time wireless data streaming.
- Led Harvard IRB-approved human-subject studies (25+ participants, 12 weight-training exercises).
- Built a deep-learning model for multi-class exercise classification and motion trajectory estimation (98%+ accuracy).
- Mentored a Harvard senior on signal processing, human-subject research, and HCRP funding applications.

Research Fellow with Prof. Conor J. Walsh

Jan. 2019 – Aug. 2020

- **Soft wearable shoulder-assist robot**

- Developed a textile-based wearable pressure array for torque estimation and accompanying capacitance readout electronics.

Korea Advanced Institute of Science and Technology (KAIST), [Biorobotics Lab](#)

Graduate Researcher with Prof. Jung Kim

Sep. 2014 – Aug. 2016

- **Skin-mountable stretch sensing with piezoresistive sensors and hysteresis compensation** [\[link\]](#)

- Developed a 3D-printed, skin-mountable soft stretch sensor.
- Ran human-subject experiments (3+ participants) to estimate multi-axis joint motion.

- **Cable-driven soft exo-glove with optical soft sensor** [\[link\]](#)

- Developed an optical soft sensor for fingertip force measurement using light-intensity modulation.
- Integrated the sensor into a cable-driven soft glove to augment grasping performance.

INDUSTRY EXPERIENCE

Wurq Inc.

Signal Processing Engineer (Part-time)

July 2022 – May 2023

- AI-powered fitness technology company
 - Developed and field-tested real-time fitness tracking algorithms for wearable hardware platforms.
 - Engineered and managed biometric data pipelines, leveraging large-scale datasets from wearable devices.
 - Built and deployed AI models for activity recognition and automated health monitoring.
 - Designed and implemented validation metrics for strength training, grounded in biomechanics and signal analysis.

Beflex Inc.

Senior Researcher

Sep. 2016 – Feb. 2018

- Biometric data-driven personal healthcare company
 - Designed and established an experimental test platform using motion capture systems to analyze biomechanical performance for personalized health metrics in runners.
 - Developed and tested a prototype with Raspberry Pi Zero microcontroller, optimizing embedded firmware and data processing algorithms for wearable devices.

PATENTS

[WO2024151781A1](#), "Methods and systems for activity detection and quantification of movement kinematics", 18 July 2024.

Inventors: D. Popov, C.J. Walsh, D. Kim, **H. Cho**, F. Bertacchi.

PUBLICATION

Journal publications

[J8] **Haedo Cho**, Zoe Kutulakos, et al., Patrick Slade. Simulation-informed human-in-the-loop optimization for multi-task with a robotic arm. In preparation, 2025.

[J7] **Haedo Cho***, Chelsey Campillo Rodriguez*, Patrick Slade. Activity-specific modes significantly improve smart-watch energy expenditure estimation. In preparation, 2025.

*These authors contributed equally.

[J6] **Haedo Cho**, Patrick Slade. A smartphone-based system for accurately estimating energy expenditure during real-world physical activity. *Nature Computational Science*, under review, 2025.

[J5] Daekyum Kim, Yichu Jin*, **Haedo Cho***, Truman Jones, Yu Meng Zhou, Ameneh Fadaie, Dmitry Popov, Krithika Swaminathan, Conor J. Walsh. Learning-based 3D human kinematics estimation using behavioral constraints from activity classification. *Nature Communications*, 16(1), 2025.

*These authors contributed equally.

[J4] Zhou, Y.M., Hohimer, C.J., Young, H.T., McCann, C.M., Pont-Esteban, D., Civici, U.S., Jin, Y., Murphy, P., Wagner, D., Cole, T., Phipps, N., **Haedo Cho**, et al. A portable inflatable soft wearable robot to assist the shoulder during industrial work. *Science Robotics*, 9(91), 2024.

[J3] Hyosang Lee*, **Haedo Cho***, Sangjoon J. Kim, Yeongjin Kim, Jung Kim. Dispenser printing of piezo-resistive nanocomposite on woven elastic fabric and hysteresis compensation for skin-mountable stretch sensing. *Smart Materials and Structures*, 27, 2018.

*These authors contributed equally.

[J2] **Haedo Cho**, Hyosang Lee, Yeongjin Kim, Jung Kim. Design of an optical soft sensor for measuring fingertip force and contact recognition. *International Journal of Control, Automation and Systems*, 15(1), 16-24, 2017.

[J1] Hyosang Lee, Donguk Kwon, **Haedo Cho**, Inkyu Park, Jung Kim. Soft nanocomposite based multi-point, multi-directional strain mapping sensor using anisotropic electrical impedance tomography. *Scientific Reports*, 7:39837, 2017.

Conference & Invited Talks

[C1] Hyosang Lee, **Jiseung Cho***, Jung Kim, “Printable skin-adhesive stretch sensor for measuring multi-axis human joint angles,” *IEEE International Conference on Robotics and Automation (ICRA)*, pp. 4975–4980, 2016. (*This paper was published before I legally changed my name.)

[C2] Yichu Jin, Christina M. Glover, **Haedo Cho**, Oluwaseun A. Araromi, Moritz A. Graule, Na Li, Robert Wood, Conor J. Walsh, “Soft sensing shirt for shoulder kinematics estimation,” *IEEE International Conference on Robotics and Automation (ICRA)*, 2020.

[C3] **Haedo Cho**, Patrick Slade, “A smartphone-based system for accurately estimating energy expenditure during real-world physical activity,” *Poster presented at the American Society of Biomechanics (ASB 2024)*, Madison, WI, August 5–8, 2024.

[T1] **Haedo Cho**, “A smartphone-based system for accurately estimating energy expenditure during real-world physical activity,” *Harvard Robotics Trainee Seminar Series*, December 2025.

TEACHING EXPERIENCE

Biomechanics of Movement and Assistive Robotics (Harvard BE 124) Fall 2023, Spring 2025
Teaching Fellow

- Undergraduate/graduate-level, 4 credits, 30 students per term.
- Lectured in lab sessions on inverse dynamics, advised final projects, and supported students during office hours.

Data Science 2: [Advanced topics in to Data Science](#) (Harvard CS109B/AC209B) Spring 2023
Teaching Fellow

- Graduate-level, 4 credits, 180 students
- Lectured in lab sessions on Gap statistics, prepared lecture slides and problem sets, graded assignments and assisted students through office hours

Data Science 1: [Introduction to Data Science](#) (Harvard CS109A/AC209A) Fall 2022
Teaching Fellow

- Graduate-level, 4 credits, 303 students
- Lectured lab session, assisted mid-term final exams led office hours

Introduction to Robotics (Harvard ES159/259) Summer-Fall 2022
Teaching Fellow

- Undergraduate/Graduate-level, 4 credits, 11 students
- Developed course materials (lecture notes and assignments), graded assignments and assisted students through office hours

Technology Venture Immersion (Harvard MS/MBA program) Fall 2021 - Winter 2022
Teaching Fellow

- Developed course materials for an introduction to IoT-based embedded systems using Arduino
- Advised Harvard MBA students on human-centered design project development

MENTORING

Truman Jones — Harvard undergraduate; now NFL player (Tennessee Titans) Feb. 2022 – Aug. 2022

- Supervised an independent capstone project fulfilling engineering degree requirements.
- Guided the student through multiple rounds of Harvard College Research Program (HCRP) funding applications.

Justin Huang — Visiting undergraduate, Northeastern University Jul. 2023 – Dec. 2025

- Mentored the development of an AWS-based platform for a smartphone-based energy expenditure system.

Filippo Andrea Mariani — Visiting M.Sc. student, Politecnico di Milano Mar. 2025 – Sep. 2025

- Advised on a master's thesis focusing on signal processing and human-subject experiments.

Matteo Amigoni — Visiting M.Sc. student, Politecnico di Milano Mar. 2025 – Sep. 2025

- Advised on a master's thesis focusing on signal processing and human-subject experiments.

HONORS AND AWARDS

Dean's Competitive Fund for Promising Scholarship Sep. 2023 – Aug. 2024
Harvard University Amount: \$50,000

Korean Government Scholarship Sep. 2014 – Aug. 2016
Ministry of Education, Science and Technology, Korea

Hanjin Scholarship Mar. 2013 – Aug. 2014
Inha University, Korea

Undergraduate Scholarship Mar. 2008 – Aug. 2012
Inha University, Korea

LEADERSHIP & SOCIAL ACTIVITIES

National Biomechanics Day [\[link\]](#) May 2025
Volunteer

- Demonstrated wearable sensing and biomechanics concepts to Boston-area high school students.

Habitat for Humanity Jul. 2013
Team Leader

- Led a volunteer team on a Habitat construction site and participated in educational activities for local children.

Republic of Korea Marine Corps Mar. 2009 – Jan. 2011
Platoon Leader

- Completed two years of mandatory military service as a platoon leader.

REFERENCES

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